

TD_Preprocessing

September 3, 2026

1 TD - Préparation et nettoyage des données avec Pandas & NumPy

1.1 Auteur

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1.2 Module

Introduction à l'IA et au Machine Learning pour les DEV

1.3 Objectif

Mise en œuvre des techniques de préparation de données :

- Manipulation de Series Pandas
- Chargement de fichiers CSV
- Gestion des valeurs manquantes
- Nettoyage des données
- Imputation avec Pandas
- Imputation avec Scikit-Learn
- Élaboration d'une stratégie de nettoyage

```
[1]: import pandas as pd
import numpy as np

from sklearn.impute import SimpleImputer
```

1.4 Exercice 1 - Manipulation d'une Series

```
[2]: sports = {
    'bask': 'BasketBall',
    'hand': 'HandBall',
    'snow': 'Snowsport',
    'base': 'BaseBall',
    'swim': 'Swimming'
}

sIndex = pd.Series(sports)
```

```
display(sIndex)
```

```
bask    Basketball
hand    HandBall
snow    Snowsport
base    BaseBall
swim    Swimming
dtype: str
```

```
[3]: # Question 1
```

```
print(sIndex.iloc[1])

print(sIndex.loc['swim'])
```

```
HandBall
Swimming
```

```
[4]: # Question 2
```

```
sIndex.str.upper()
```

```
[4]: bask    BASKETBALL
hand    HANDBALL
snow    SNOWSPORT
base    BASEBALL
swim    SWIMMING
dtype: str
```

```
[5]: # Question 3
```

```
resultat = []

for sport in sIndex:
    resultat.append(sport.upper())

resultat
```

```
[5]: ['BASKETBALL', 'HANDBALL', 'SNOWSPORT', 'BASEBALL', 'SWIMMING']
```

```
[8]: # Question 4
```

```
%timeit sIndex.str.upper()
```

76.9 s ± 2.44 s per loop (mean ± std. dev. of 7 runs, 10,000 loops each)

1.4.1 Conclusion

La solution Pandas est plus rapide grâce à la vectorisation.

1.5 Exercice 2 - Chargement d'un CSV

```
[9]: Clients_Data = pd.read_csv("../data/Custemers.csv")
```

```
[10]: Clients_Data.shape
```

```
[10]: (10000, 15)
```

```
[11]: Clients_Data.head()
```

```
[11]:
```

	name \
0	Deonte Stark
1	Faustino Boyer
2	Eddy Bogisich,33431 Dollie Squares Apt. 654,Po...
3	Mervyn Kreiger
4	Katlyn Doyle,4650 Beer Crossing Suite 848,Nort...

	address	city	state	zip \
0	278 Mueller Plains	North Euna	Alabama	03404-4384
1	70244 Skiles Falls Suite 030	North Altohaven	California	01522-1310
2	NaN	NaN	NaN	NaN
3	376 Dorinda Stream	Shaniquafort	South Carolina	39347-4438
4	NaN	NaN	NaN	NaN

	phone	email	work \
0	(180)940-9676x4495	shanna73@hotmail.com	Hahn-Mayer
1	(308)699-6239x81011	ferrell81@gmail.com	Buckridge Inc
2	NaN	NaN	NaN
3	869-985-6299	emmerich.griselda@hotmail.com	Witting PLC
4	NaN	NaN	NaN

	work address	work city	work state	work zipcode \
0	8177 Weber Throughway Apt. 341	Jaronton	Maine	51589-1424
1	236 Kessler Center	New Gavynshire	Missouri	52234-6972
2	NaN	NaN	NaN	NaN
3	521 Kemmer Manors	Nerytown	Kentucky	68774
4	NaN	NaN	NaN	NaN

	work phone	work email	account created on
0	01240240340	heller.kirstin@glover.com	2001-09-06 06:15:24
1	(486)896-6855x446	esta.dicki@bechtelar.com	1983-04-25
2	NaN	NaN	NaN
3	(212)169-8190	greyson39@purdy.com	1971-04-27 14:05:06
4	NaN	NaN	NaN

```
[60]: Clients_Data.columns
```

```
[60]: Index(['name', 'address', 'city', 'state', 'zip', 'phone', 'email', 'work',  
          'work address', 'work city', 'work state', 'work zipcode', 'work phone',  
          'work email', 'account created on', 'firstname', 'lastname'],  
          dtype='str')
```

1.6 Exercice 3 - Détection des valeurs manquantes

```
[13]: Clients_Data.isnull().sum()
```

```
[13]: name                81  
      address            3365  
      city              3374  
      state             3365  
      zip               3362  
      phone             3366  
      email             3363  
      work              3390  
      work address      3383  
      work city         3378  
      work state        3390  
      work zipcode      3378  
      work phone        3377  
      work email        3359  
      account created on 3368  
      dtype: int64
```

```
[14]: Clients_Data.isnull().mean()*100
```

```
[14]: name                0.81  
      address            33.65  
      city              33.74  
      state             33.65  
      zip               33.62  
      phone             33.66  
      email             33.63  
      work              33.90  
      work address      33.83  
      work city         33.78  
      work state        33.90  
      work zipcode      33.78  
      work phone        33.77  
      work email        33.59  
      account created on 33.68  
      dtype: float64
```

```
[15]: Clients_Data[  
      Clients_Data['name'].isnull()
```

]

```
[15]:      name                address                city                state \
148   NaN      558 Brycen Mission Suite 152      Cristmouth      Arkansas
273   NaN      0481 Sanford Lake Apt. 439      Bashirianberg North Carolina
300   NaN      38689 Kimora Groves Suite 807      New Nadiahaven      Vermont
330   NaN      077 Walsh Summit Suite 123      Rogahnfurt      Indiana
467   NaN      87140 Loma Crescent      North Dixieport      Michigan
...   ...
9316  NaN      054 Aubrie Corners      East Genevieve      New Hampshire
9487  NaN      9957 Rempel Wells Apt. 081      New Micky      Alabama
9745  NaN      6277 Schneider Common Suite 939      Port Aura      North Dakota
9816  NaN      986 Brianne Shoals      Port Wilbertstad      Mississippi
9834  NaN      19233 Kreiger Meadow      Kleinville      Michigan

      zip                phone                email \
148   52585-7480      +03(6)2449148729      corene08@gmail.com
273   61948-9865      (967)713-7747x7329      hermiston.jenniffer@hotmail.com
300   99133-2546      595.341.7775x50639      sfritsch@yahoo.com
330   81369      +38(1)3931574807      tdenesik@hotmail.com
467   32196-8397      +65(0)2946604451      lucille69@gmail.com
...   ...
9316   92371      296-242-9406x586      enrico.kuvalis@yahoo.com
9487   24013      522-665-0735      rylan47@hotmail.com
9745   10127      127.858.7550      kacy29@gmail.com
9816   99396      953.475.7297x373      ritchie.hadley@yahoo.com
9834  42537-0382      (894)710-7706x071      hjalmer14@hotmail.com

      work                work address                work city \
148      Braun PLC      8795 Carissa Land Apt. 884      South Arachester
273      Grant LLC      7375 Lynn Fork      Wuckerthaven
300      Purdy-Farrell      29365 Nyah Flats      West Isabella
330      Corwin-Fahey      717 Karol Stravenue Apt. 964      Trantowhaven
467      Monahan Ltd      99220 Murphy Motorway      East Reyhaven
...   ...
9316      Schneider LLC      53056 Gaige Motorway      North Audriana
9487      Bergnaum and Sons      60120 Klocko Alley Apt. 615      Adalineport
9745      Becker PLC      186 Armstrong Lakes      East Rosalyn
9816      Kautzer-Batz      22243 Feil Terrace Suite 469      New Catofurt
9834      Beer-Ledner      48801 Bobbye Roads      West Tylor

      work state work zipcode                work phone \
148      New Mexico      68070-4086      09359769734
273      Ohio      89342-4989      419-132-5389
300      Florida      06342      +66(4)7117753075
330      Utah      49873-6641      882.577.0021x2495
467      Rhode Island      50641      773.245.8906
```

...	...	...	...
9316	Arkansas	73816	605.101.0879
9487	Maryland	37182-8209	289.393.5802x9949
9745	Ohio	07576	1-924-436-1559x850
9816	Wisconsin	62178	09723145605
9834	Arkansas	60338	849.567.9201x682

	work email	account created on
148	hezzie.hettinger@mcglynnlarson.com	1999-12-12
273	colin.walsh@hane.com	1995-08-07 19:52:30
300	ocrooks@mohr.net	1975-04-07
330	mschaden@gerholdschultz.net	03/10/2010
467	parisian.willard@feil.org	04/29/1986

...	...	...
9316	cephus.conroy@schroeder.com	01/21/2000
9487	mwolf@corwin.net	2000-09-07
9745	kunde.carry@auerweimann.org	1995-08-13 11:03:45
9816	thompson.eulalie@mann.com	1970-01-27 16:26:56
9834	hsatterfield@bradtke.com	2015-10-13 23:24:40

[81 rows x 15 columns]

```
[16]: Clients_Data.dropna(
      how="all"
      )
```

```
[16]:
```

	name \
0	Deonte Stark
1	Faustino Boyer
2	Eddy Bogisich,33431 Dollie Squares Apt. 654,Po...
3	Mervyn Kreiger
4	Katlyn Doyle,4650 Beer Crossing Suite 848,Nort...
...	...
9995	Ciera Wunsch
9996	Beula Bauch
9997	Garvin Runolfsson
9998	Hildegarde Gleichner
9999	Rhett Spinka,87035 Solon Prairie,Zelmahaven,Ha...

	address	city	state \
0	278 Mueller Plains	North Euna	Alabama
1	70244 Skiles Falls Suite 030	North Altohaven	California
2	NaN	NaN	NaN
3	376 Dorinda Stream	Shaniquafort	South Carolina
4	NaN	NaN	NaN
...	...	...	...
9995	25198 Fernando Isle Suite 175	South Pricillashire	Colorado

9996	800 Keebler Stream	East Denise	Maine
9997	4844 Dietrich Islands Suite 457	Auerport	Michigan
9998	96841 Predovic Locks	New Knute	Rhode Island
9999	NaN	NaN	NaN

	zip	phone	email \
0	03404-4384	(180)940-9676x4495	shanna73@hotmail.com
1	01522-1310	(308)699-6239x81011	ferrell181@gmail.com
2	NaN	NaN	NaN
3	39347-4438	869-985-6299	emmerich.griselda@hotmail.com
4	NaN	NaN	NaN
...	...	...	...
9995	53222-8926	005-769-2290x3662	shannon.beatty@gmail.com
9996	33693	933.047.0305	steuber.zakary@hotmail.com
9997	34497	312.046.9856	nwehner@hotmail.com
9998	88719-7033	269-922-1275	ksatterfield@yahoo.com
9999	NaN	NaN	NaN

	work	work address	work city \
0	Hahn-Mayer	8177 Weber Throughway Apt. 341	Jaronton
1	Buckridge Inc	236 Kessler Center	New Gavynshire
2	NaN	NaN	NaN
3	Witting PLC	521 Kemmer Manors	Nerytown
4	NaN	NaN	NaN
...	...	...	...
9995	Metz-Kub	28254 Nikolaus Plains	West Emmithaven
9996	Barrows Inc	35083 Champlin Keys Suite 030	Port Yulisa
9997	Braun Inc	7514 Samiyah Brooks	West Ham
9998	Turcotte LLC	NaN	Medhurstside
9999	NaN	NaN	NaN

	work state	work zipcode	work phone \
0	Maine	51589-1424	01240240340
1	Missouri	52234-6972	(486)896-6855x446
2	NaN	NaN	NaN
3	Kentucky	68774	(212)169-8190
4	NaN	NaN	NaN
...	...	...	...
9995	Minnesota	34939-5981	(154)316-2907x520
9996	Rhode Island	35745	(148)771-5153x521
9997	Texas	90804	388-879-8433x7099
9998	Oregon	20523-8984	863.095.3340x2593
9999	NaN	NaN	NaN

	work email	account created on
0	heller.kirstin@glover.com	2001-09-06 06:15:24
1	esta.dicki@bechtelar.com	1983-04-25

2		NaN	NaN
3	greyson39@purdy.com	1971-04-27	14:05:06
4		NaN	NaN
...	...	...	...
9995	clemente.dach@leannon.com		1986-11-14
9996	windler.trudi@shanahanhagenes.org	2013-11-02	20:50:10
9997	stamm.erlene@stoltenberg.org		2008-04-13
9998	coby06@kulasbogan.com		1990-02-20
9999		NaN	NaN

[10000 rows x 15 columns]

```
[17]: Clients_Data.dropna(
      axis=0
      )
```

```
[17]:
```

	name	address \
0	Deonte Stark	278 Mueller Plains
1	Faustino Boyer	70244 Skiles Falls Suite 030
3	Mervyn Kreiger	376 Dorinda Stream
5	Daquan Leffler	43142 Howell Ports Apt. 953
6	Jace Konopelski	963 Senger Points Apt. 992
...	...	...
9991	Tawnya Williamson	4160 Welch Roads
9993	Cayla Gleason	505 Rogahn Trafficway
9995	Ciera Wunsch	25198 Fernando Isle Suite 175
9996	Beula Bauch	800 Keebler Stream
9997	Garvin Runolfsson	4844 Dietrich Islands Suite 457

	city	state	zip	phone \
0	North Euna	Alabama	03404-4384	(180)940-9676x4495
1	North Altohaven	California	01522-1310	(308)699-6239x81011
3	Shaniquafort	South Carolina	39347-4438	869-985-6299
5	West Emory	Connecticut	08086-5571	874-576-0739
6	Lake Fonda	North Dakota	15581	(083)182-4626x8012
...	...	...	...	...
9991	Amasahaven	Maine	76161	1-468-320-0246x8915
9993	New Sabastian	Tennessee	56989-4233	(578)118-9244x85865
9995	South Pricillashire	Colorado	53222-8926	005-769-2290x3662
9996	East Denise	Maine	33693	933.047.0305
9997	Auerport	Michigan	34497	312.046.9856

	email	work \
0	shanna73@hotmail.com	Hahn-Mayer
1	ferrell81@gmail.com	Buckridge Inc
3	emmerich.griselda@hotmail.com	Witting PLC
5	natosha.kutch@yahoo.com	Christiansen Ltd

6	langosh.mathilde@hotmail.com	Glover-Howe
...	...	...
9991	doreilly@yahoo.com	Schuster-Lemke
9993	mtorp@gmail.com	Morissette Group
9995	shannon.beatty@gmail.com	Metz-Kub
9996	steuber.zakary@hotmail.com	Barrows Inc
9997	nwehner@hotmail.com	Braun Inc

	work address	work city	work state \
0	8177 Weber Throughway Apt. 341	Jaronton	Maine
1	236 Kessler Center	New Gavynshire	Missouri
3	521 Kemmer Manors	Nerytown	Kentucky
5	3136 Jered Junction Suite 993	West Mallie	New Hampshire
6	87986 Harvey Islands	South Zackary	Kentucky
...	...	...	...
9991	82853 Watsica View	South Jaydan	Georgia
9993	484 Lubowitz Inlet Suite 138	East Sannieland	North Carolina
9995	28254 Nikolaus Plains	West Emmithaven	Minnesota
9996	35083 Champlin Keys Suite 030	Port Yulisa	Rhode Island
9997	7514 Samiyah Brooks	West Ham	Texas

	work zipcode	work phone	work email \
0	51589-1424	01240240340	heller.kirstin@glover.com
1	52234-6972	(486)896-6855x446	esta.dicki@bechtelar.com
3	68774	(212)169-8190	greyson39@purdy.com
5	40778-7891	215.605.7824x844	anderson.gifford@nader.com
6	75232-9391	(270)125-2639	kassulke.fate@hills.com
...	...	...	...
9991	54467-3704	264-477-5271	donell195@wilderman.com
9993	66018	892-930-2586x31401	bina.pagac@johns.com
9995	34939-5981	(154)316-2907x520	clemente.dach@leannon.com
9996	35745	(148)771-5153x521	windler.trudi@shanahanhagenes.org
9997	90804	388-879-8433x7099	stamm.erlene@stoltenberg.org

	account created on
0	2001-09-06 06:15:24
1	1983-04-25
3	1971-04-27 14:05:06
5	2014-01-15 23:15:43
6	05/20/2004
...	...
9991	1974-07-22
9993	1996-04-08
9995	1986-11-14
9996	2013-11-02 20:50:10
9997	2008-04-13

```
[5657 rows x 15 columns]
```

1.6.1 Observation

La suppression directe des lignes peut entraîner une perte importante d'informations.

1.7 Exercice 4 - Vérification des types et nettoyage

```
[62]: print(Clients_Data.dtypes)
```

```
name                str
address             str
city                str
state               str
zip                 str
phone               str
email               str
work                str
work address        str
work city            str
work state          str
work zipcode        str
work phone           str
work email          str
account created on  str
firstname            str
lastname             str
dtype: object
```

```
[18]: Clients_Data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   name                  9919 non-null  str
1   address               6635 non-null  str
2   city                  6626 non-null  str
3   state                 6635 non-null  str
4   zip                   6638 non-null  str
5   phone                 6634 non-null  str
6   email                 6637 non-null  str
7   work                  6610 non-null  str
8   work address          6617 non-null  str
9   work city             6622 non-null  str
10  work state            6610 non-null  str
11  work zipcode          6622 non-null  str
12  work phone            6623 non-null  str
```

```

13 work email          6641 non-null  str
14 account created on  6632 non-null  str
dtypes: str(15)
memory usage: 1.1 MB

```

```

[19]: Clients_Data[
      ["firstname","lastname"]
] = (
      Clients_Data["name"]
      .str.split(
          " ",
          n=1,
          expand=True
      )
)

```

```

[20]: Clients_Data.head()

```

```

[20]:

```

	name \
0	Deonte Stark
1	Faustino Boyer
2	Eddy Bogisich,33431 Dollie Squares Apt. 654,Po...
3	Mervyn Kreiger
4	Katlyn Doyle,4650 Beer Crossing Suite 848,Nort...

	address	city	state	zip \
0	278 Mueller Plains	North Euna	Alabama	03404-4384
1	70244 Skiles Falls Suite 030	North Altohaven	California	01522-1310
2	NaN	NaN	NaN	NaN
3	376 Dorinda Stream	Shaniquafort	South Carolina	39347-4438
4	NaN	NaN	NaN	NaN

	phone	email	work \
0	(180)940-9676x4495	shanna73@hotmail.com	Hahn-Mayer
1	(308)699-6239x81011	ferrell181@gmail.com	Buckridge Inc
2	NaN	NaN	NaN
3	869-985-6299	emmerich.griselda@hotmail.com	Witting PLC
4	NaN	NaN	NaN

	work address	work city	work state	work zipcode \
0	8177 Weber Throughway Apt. 341	Jaronton	Maine	51589-1424
1	236 Kessler Center	New Gavynshire	Missouri	52234-6972
2	NaN	NaN	NaN	NaN
3	521 Kemmer Manors	Nerytown	Kentucky	68774
4	NaN	NaN	NaN	NaN

	work phone	work email	account created on \
--	------------	------------	----------------------

```

0      01240240340 heller.kirstin@glover.com 2001-09-06 06:15:24
1 (486)896-6855x446 esta.dicki@bechtelar.com 1983-04-25
2      NaN NaN NaN
3 (212)169-8190 greyson39@purdy.com 1971-04-27 14:05:06
4      NaN NaN NaN

```

```

      firstname lastname
0 Deonte Stark
1 Faustino Boyer
2 Eddy Bogisich,33431 Dollie Squares Apt. 654,Port Ca...
3 Mervyn Kreiger
4 Katlyn Doyle,4650 Beer Crossing Suite 848,North Dells...

```

```
[64]: Clients_Data["lastname"].str.len()
```

```

[64]: 2      239
      4      239
      9      223
     10      271
     13      213
      ...
    9987      240
    9994      249
    9997       17
    9998       20
    9999      245
Name: lastname, Length: 3627, dtype: int64

```

```
[68]: Clients_Data = Clients_Data[
      Clients_Data["lastname"].str.len() > 16
      ]
```

```
[70]: print(
      Clients_Data
      )
```

```

                                     name \
2      Eddy Bogisich,33431 Dollie Squares Apt. 654,Po...
4      Katlyn Doyle,4650 Beer Crossing Suite 848,Nort...
9      Weston Ebert,03552 Lehner Ridge,New Attilio,Ha...
10     Joshua Mosciski,31085 Schmitt Parkways Apt. 9...
13     Jovanny Armstrong,7334 Kris Flat,,Florida,792...
      ...
    9987 Mervyn Krajcik,34259 Pagac Pike,Schmittland,Ok...
    9994 Wiley Bradtke,9508 Simonis Crossing Suite 502,...
    9997                                     Garvin Runolfsson
    9998                                     Hildegard Gleichner
    9999 Rhett Spinka,87035 Solon Prairie,Zelmahaven,Ha...

```

	address	city	state	zip \
2	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN
9	NaN	NaN	NaN	NaN
10	NaN	NaN	NaN	NaN
13	NaN	NaN	NaN	NaN
...	...	...	...	...
9987	NaN	NaN	NaN	NaN
9994	NaN	NaN	NaN	NaN
9997	4844 Dietrich Islands Suite 457	Auerport	Michigan	34497
9998	96841 Predovic Locks	New Knute	Rhode Island	88719-7033
9999	NaN	NaN	NaN	NaN

	phone	email	work	work address \
2	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN
9	NaN	NaN	NaN	NaN
10	NaN	NaN	NaN	NaN
13	NaN	NaN	NaN	NaN
...	...	...	...	...
9987	NaN	NaN	NaN	NaN
9994	NaN	NaN	NaN	NaN
9997	312.046.9856	nwehner@hotmail.com	Braun Inc	7514 Samiyah Brooks
9998	269-922-1275	ksatterfield@yahoo.com	Turcotte LLC	NaN
9999	NaN	NaN	NaN	NaN

	work city	work state	work zipcode	work phone \
2	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN
9	NaN	NaN	NaN	NaN
10	NaN	NaN	NaN	NaN
13	NaN	NaN	NaN	NaN
...	...	...	...	...
9987	NaN	NaN	NaN	NaN
9994	NaN	NaN	NaN	NaN
9997	West Ham	Texas	90804	388-879-8433x7099
9998	Medhurstside	Oregon	20523-8984	863.095.3340x2593
9999	NaN	NaN	NaN	NaN

	work email	account	created on	firstname \
2	NaN	NaN	NaN	Eddy
4	NaN	NaN	NaN	Katlyn
9	NaN	NaN	NaN	Weston
10	NaN	NaN	NaN	Joshuah
13	NaN	NaN	NaN	
...	...	...	...	...
9987	NaN	NaN	NaN	Mervyn

9994		NaN	NaN	Wiley
9997	stamm.erlene@stoltenberg.org		2008-04-13	
9998	coby06@kulasbogan.com		1990-02-20	
9999		NaN	NaN	Rhett

			lastname
2	Bogisich,33431 Dollie Squares Apt. 654,Port Ca...		
4	Doyle,4650 Beer Crossing Suite 848,North Dells...		
9	Ebert,03552 Lehner Ridge,New Attilio,Hawaii,35...		
10	Mosciski,31085 Schmitt Parkways Apt. 957,Dibbe...		
13	Jovanny Armstrong,7334 Kris Flat,,Florida,7922...		
...			...
9987	Krajcik,34259 Pagac Pike,Schmittland,Oklahoma,...		
9994	Bradtke,9508 Simonis Crossing Suite 502,East H...		
9997		Garvin Runolfsson	
9998		Hildegarde Gleichner	
9999	Spinka,87035 Solon Prairie,Zelmahaven,Hawaii,3...		

[3627 rows x 17 columns]

1.8 Exercice 5 — Imputation pour traiter les données manquantes

```
[23]: origin_Data = pd.read_csv(
    "../data/melb_data.csv"
)
```

```
[24]: origin_Data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 18396 entries, 0 to 18395
Data columns (total 22 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      18396 non-null  int64
1   Suburb          18396 non-null  str
2   Address         18396 non-null  str
3   Rooms           18396 non-null  int64
4   Type            18396 non-null  str
5   Price           18396 non-null  float64
6   Method          18396 non-null  str
7   SellerG         18396 non-null  str
8   Date            18396 non-null  str
9   Distance        18395 non-null  float64
10  Postcode        18395 non-null  float64
11  Bedroom2        14927 non-null  float64
12  Bathroom        14925 non-null  float64
13  Car              14820 non-null  float64
14  Landsize        13603 non-null  float64
```

```

15 BuildingArea    7762 non-null    float64
16 YearBuilt      8958 non-null    float64
17 CouncilArea    12233 non-null    str
18 Lattitude      15064 non-null   float64
19 Longitude      15064 non-null   float64
20 Regionname     18395 non-null    str
21 Propertycount  18395 non-null    float64
dtypes: float64(12), int64(2), str(8)
memory usage: 3.1 MB

```

```
[25]: origin_Data.shape
```

```
[25]: (18396, 22)
```

```
[26]: origin_Data.isnull().sum()
```

```

[26]: Unnamed: 0      0
      Suburb        0
      Address       0
      Rooms         0
      Type          0
      Price         0
      Method        0
      SellerG       0
      Date          0
      Distance      1
      Postcode      1
      Bedroom2     3469
      Bathroom     3471
      Car          3576
      Landsize     4793
      BuildingArea 10634
      YearBuilt    9438
      CouncilArea  6163
      Lattitude    3332
      Longitude    3332
      Regionname   1
      Propertycount 1
      dtype: int64

```

```
[27]: origin_Data.isnull().sum(axis=1)
```

```

[27]: 0      2
      1      0
      2      0
      3      2
      4      0
      ..

```

```
18391    2
18392    1
18393    4
18394    1
18395    2
Length: 18396, dtype: int64
```

```
[28]: def missing_columns(originDB):

    colsWithMissing = [
        col
        for col in originDB.columns
        if originDB[col].isnull().any()
    ]

    reduced_original_data = originDB.drop(
        colsWithMissing,
        axis=1
    )

    return colsWithMissing, reduced_original_data
```

```
[29]: colsWithMissing, reducedData = (
        missing_columns(origin_Data)
    )

    colsWithMissing
```

```
[29]: ['Distance',
        'Postcode',
        'Bedroom2',
        'Bathroom',
        'Car',
        'Landsize',
        'BuildingArea',
        'YearBuilt',
        'CouncilArea',
        'Latitude',
        'Longitude',
        'Regionname',
        'Propertycount']
```

```
[30]: origin_Data.isnull().mean()*100
```

```
[30]: Unnamed: 0      0.000000
      Suburb        0.000000
      Address       0.000000
```

```

Rooms          0.000000
Type           0.000000
Price          0.000000
Method         0.000000
SellerG        0.000000
Date           0.000000
Distance       0.005436
Postcode       0.005436
Bedroom2       18.857360
Bathroom       18.868232
Car            19.439008
Landsize       26.054577
BuildingArea   57.806045
YearBuilt      51.304631
CouncilArea    33.501848
Lattitude      18.112633
Longitude      18.112633
Regionname     0.005436
Propertycount  0.005436
dtype: float64

```

```
[31]: origin_Data.isnull().mean(axis=1)*100
```

```

[31]: 0          9.090909
      1          0.000000
      2          0.000000
      3          9.090909
      4          0.000000
      ...
      18391      9.090909
      18392      4.545455
      18393      18.181818
      18394      4.545455
      18395      9.090909
      Length: 18396, dtype: float64

```

```
[32]: new_Data = origin_Data[
      origin_Data.isnull().mean(axis=1)*100 <= 5
      ]
```

```
[69]: print(
      new_Data
      )
```

	Unnamed: 0	Suburb	Address	Rooms	Type	Price \
1	2	Abbotsford	25 Bloomburg St	2	h	1035000.0
2	4	Abbotsford	5 Charles St	3	h	1465000.0
4	6	Abbotsford	55a Park St	4	h	1600000.0

6	11	Abbotsford	124 Yarra St	3	h	1876000.0
7	14	Abbotsford	98 Charles St	2	h	1636000.0
...	...	...	...	...	...	...
18382	23527	Watsonia	76 Kenmare St	2	h	650000.0
18384	23532	Werribee	5 Nuragi Ct	4	h	635000.0
18389	23538	Williamstown	77 Merrett Dr	3	h	1031000.0
18392	23541	Williamstown	96 Verdon St	4	h	2500000.0
18394	23545	Yarraville	6 Agnes St	4	h	1285000.0

	Method	SellerG	Date	Distance	...	Bathroom	Car	\
1	S	Biggin	4/02/2016	2.5	...	1.0	0.0	
2	SP	Biggin	4/03/2017	2.5	...	2.0	0.0	
4	VB	Nelson	4/06/2016	2.5	...	1.0	2.0	
6	S	Nelson	7/05/2016	2.5	...	2.0	0.0	
7	S	Nelson	8/10/2016	2.5	...	1.0	2.0	

...	...	...	...	...	...	...	...	...
18382	PI	Morrison	26/08/2017	14.5	...	1.0	1.0	
18384	S	hockingstuart	26/08/2017	14.7	...	2.0	1.0	
18389	SP	Williams	26/08/2017	6.8	...	2.0	2.0	
18392	PI	Sweeney	26/08/2017	6.8	...	1.0	5.0	
18394	SP	Village	26/08/2017	6.3	...	1.0	1.0	

	Landsize	BuildingArea	YearBuilt	CouncilArea	Lattitude	Longitude	\
1	156.0	79.0	1900.0	Yarra	-37.80790	144.99340	
2	134.0	150.0	1900.0	Yarra	-37.80930	144.99440	
4	120.0	142.0	2014.0	Yarra	-37.80720	144.99410	
6	245.0	210.0	1910.0	Yarra	-37.80240	144.99930	
7	256.0	107.0	1890.0	Yarra	-37.80600	144.99540	
...	...	...	...	...	...	...	...
18382	210.0	79.0	2006.0	NaN	-37.70657	145.07878	
18384	662.0	172.0	1980.0	NaN	-37.89327	144.64789	
18389	333.0	133.0	1995.0	NaN	-37.85927	144.87904	
18392	866.0	157.0	1920.0	NaN	-37.85908	144.89299	
18394	362.0	112.0	1920.0	NaN	-37.81188	144.88449	

	Regionname	Propertycount
1	Northern Metropolitan	4019.0
2	Northern Metropolitan	4019.0
4	Northern Metropolitan	4019.0
6	Northern Metropolitan	4019.0
7	Northern Metropolitan	4019.0
...	...	...
18382	Northern Metropolitan	2329.0
18384	Western Metropolitan	16166.0
18389	Western Metropolitan	6380.0
18392	Western Metropolitan	6380.0
18394	Western Metropolitan	6543.0

[8332 rows x 22 columns]

1.9 Exercice 6 - Imputation avec SimpleImputer

```
[33]: origin_Data_num = (  
        origin_Data.select_dtypes(  
            include=['number']  
        )  
    )  
  
[34]: imputer = SimpleImputer(  
        strategy='mean'  
    )  
  
[35]: imputer.fit(origin_Data_num)  
  
[35]: SimpleImputer()  
  
[36]: imputer.statistics_  
  
[36]: array([ 1.18267871e+04,  2.93504023e+00,  1.05669746e+06,  1.03899864e+01,  
            3.10714015e+03,  2.91304348e+00,  1.53849246e+00,  1.61551957e+00,  
            5.58116371e+02,  1.51220219e+02,  1.96588000e+03, -3.78098491e+01,  
            1.44996338e+02,  7.51797527e+03])
```

```
[37]: transform_data = (  
        imputer.transform(  
            origin_Data_num  
        )  
    )
```

```
[38]: df_transform = pd.DataFrame(  
        transform_data,  
        columns=origin_Data_num.columns,  
        index=origin_Data_num.index  
    )
```

```
[39]: df_transform.head()
```

```
[39]:   Unnamed: 0  Rooms      Price  Distance  Postcode  Bedroom2  Bathroom  Car  \  
0         1.0    2.0  1480000.0         2.5    3067.0         2.0     1.0  1.0  
1         2.0    2.0  1035000.0         2.5    3067.0         2.0     1.0  0.0  
2         4.0    3.0  1465000.0         2.5    3067.0         3.0     2.0  0.0  
3         5.0    3.0   850000.0         2.5    3067.0         3.0     2.0  1.0  
4         6.0    4.0  1600000.0         2.5    3067.0         3.0     1.0  2.0  
  
   Landsize  BuildingArea  YearBuilt  Lattitude  Longtitude  Propertycount  
0     202.0    151.220219  1965.879996   -37.7996    144.9984         4019.0
```

1	156.0	79.000000	1900.000000	-37.8079	144.9934	4019.0
2	134.0	150.000000	1900.000000	-37.8093	144.9944	4019.0
3	94.0	151.220219	1965.879996	-37.7969	144.9969	4019.0
4	120.0	142.000000	2014.000000	-37.8072	144.9941	4019.0

1.10 Exercice 7 - Nettoyage du dataset Olympics

```
[40]: olympics = pd.read_csv(
      ..../data/olympics.csv"
    )
```

```
[41]: olympics.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 16 columns):
#   Column  Non-Null Count  Dtype
---  -
0   0        147 non-null        str
1   1        148 non-null        str
2   2        148 non-null        str
3   3        148 non-null        str
4   4        148 non-null        str
5   5        148 non-null        str
6   6        148 non-null        str
7   7        148 non-null        str
8   8        148 non-null        str
9   9        148 non-null        str
10  10       148 non-null        str
11  11       148 non-null        str
12  12       148 non-null        str
13  13       148 non-null        str
14  14       148 non-null        str
15  15       148 non-null        str
dtypes: str(16)
memory usage: 18.6 KB
```

```
[42]: olympics.shape
```

```
[42]: (148, 16)
```

```
[43]: olympics.head()
```

```
[43]:
```

	0	1	2	3	4	5	6	7	8	\
0	NaN	№ Summer	01 !	02 !	03 !	Total	№ Winter	01 !	02 !	
1	Afghanistan (AFG)	13	0	0	2	2	0	0	0	
2	Algeria (ALG)	12	5	2	8	15	3	0	0	
3	Argentina (ARG)	23	18	24	28	70	18	0	0	

4	Armenia (ARM)			5	1	2	9	12	6	0	0
	9	10	11	12	13	14	15				
0	03 !	Total	№ Games	01 !	02 !	03 !	Combined total				
1	0	0	13	0	0	2	2				
2	0	0	15	5	2	8	15				
3	0	0	41	18	24	28	70				
4	0	0	11	1	2	9	12				

```
[44]: olympics.isnull().sum()
```

```
[44]: 0      1
      1      0
      2      0
      3      0
      4      0
      5      0
      6      0
      7      0
      8      0
      9      0
     10      0
     11      0
     12      0
     13      0
     14      0
     15      0
      dtype: int64
```

```
[46]: olympics.duplicated().sum()
```

```
[46]: np.int64(0)
```

```
[47]: olympics.drop_duplicates(
      inplace=True
      )
```

```
[48]: olympics.dropna(
      how="all",
      inplace=True
      )
```

1.10.1 Stratégie appliquée

- Vérification de la structure
- Vérification des types
- Détection des valeurs manquantes
- Suppression des doublons

- Suppression des lignes totalement vides

1.11 Exercice 7 - Nettoyage du dataset Flickr

```
[49]: flicker = pd.read_csv(
      ..../data/flicker.csv"
    )
```

```
[50]: flicker.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 8287 entries, 0 to 8286
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Identifier                            8287 non-null   int64
1   Edition Statement                    773 non-null    str
2   Place of Publication                 8287 non-null   str
3   Date of Publication                  8106 non-null   str
4   Publisher                            4092 non-null   str
5   Title                                8287 non-null   str
6   Author                              6509 non-null   str
7   Contributors                         8287 non-null   str
8   Corporate Author                    0 non-null      float64
9   Corporate Contributors              0 non-null      float64
10  Former owner                         1 non-null      str
11  Engraver                             0 non-null      float64
12  Issuance type                        8287 non-null   str
13  Flickr URL                           8287 non-null   str
14  Shelfmarks                           8287 non-null   str
dtypes: float64(3), int64(1), str(11)
memory usage: 971.3 KB
```

```
[51]: flicker.shape
```

```
[51]: (8287, 15)
```

```
[52]: flicker.head()
```

```
[52]:
```

	Identifier	Edition Statement	Place of Publication \
0	206	NaN	London
1	216	NaN	London; Virtue & Yorston
2	218	NaN	London
3	472	NaN	London
4	480	A new edition, revised, etc.	London

	Date of Publication	Publisher \
0	1879 [1878]	S. Tinsley & Co.

1	1868	Virtue & Co.
2	1869	Bradbury, Evans & Co.
3	1851	James Darling
4	1857	Wertheim & Macintosh

	Title	Author \
0	Walter Forbes. [A novel.] By A. A	A. A.
1	All for Greed. [A novel. The dedication signed...	A., A. A.
2	Love the Avenger. By the author of "All for Gr...	A., A. A.
3	Welsh Sketches, chiefly ecclesiastical, to the...	A., E. S.
4	[The World in which I live, and my place in it...	A., E. S.

	Contributors	Corporate Author \
0	FORBES, Walter.	NaN
1	BLAZE DE BURY, Marie Pauline Rose - Baroness	NaN
2	BLAZE DE BURY, Marie Pauline Rose - Baroness	NaN
3	Appleyard, Ernest Silvanus.	NaN
4	BROOME, John Henry.	NaN

	Corporate Contributors	Former owner	Engraver	Issuance type \
0	NaN	NaN	NaN	monographic
1	NaN	NaN	NaN	monographic
2	NaN	NaN	NaN	monographic
3	NaN	NaN	NaN	monographic
4	NaN	NaN	NaN	monographic

	Flickr URL \
0	http://www.flickr.com/photos/britishlibrary/ta...
1	http://www.flickr.com/photos/britishlibrary/ta...
2	http://www.flickr.com/photos/britishlibrary/ta...
3	http://www.flickr.com/photos/britishlibrary/ta...
4	http://www.flickr.com/photos/britishlibrary/ta...

	Shelfmarks
0	British Library HMNTS 12641.b.30.
1	British Library HMNTS 12626.cc.2.
2	British Library HMNTS 12625.dd.1.
3	British Library HMNTS 10369.bbb.15.
4	British Library HMNTS 9007.d.28.

```
[53]: flicker.isnull().sum()
```

```
[53]: Identifier          0
      Edition Statement    7514
      Place of Publication 0
      Date of Publication   181
      Publisher            4195
```

Title	0
Author	1778
Contributors	0
Corporate Author	8287
Corporate Contributors	8287
Former owner	8286
Engraver	8287
Issuance type	0
Flickr URL	0
Shelfmarks	0
dtype:	int64

```
[54]: flicker.duplicated().sum()
```

```
[54]: np.int64(0)
```

```
[55]: flicker.dropna(
        how="all",
        inplace=True
    )
```

```
[57]: num_cols = flicker.select_dtypes(
        include="number"
    )

num_cols = num_cols.dropna(
    axis=1,
    how="all"
)

imputer = SimpleImputer(
    strategy="mean"
)

result = imputer.fit_transform(
    num_cols
)

num_cols_clean = pd.DataFrame(
    result,
    columns=num_cols.columns,
    index=num_cols.index
)

num_cols_clean.head()
```

```
[57]: Identifier
0      206.0
1      216.0
2      218.0
3      472.0
4      480.0
```

1.11.1 Conclusion générale

Les datasets ont été explorés, nettoyés et préparés pour être utilisés dans un pipeline de Machine Learning.

Les principales étapes ont été :

- Analyse exploratoire
- Détection des valeurs manquantes
- Nettoyage
- Imputation
- Vérification finale de la qualité des données